

Generation through the lens of learning theory

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Growing interest in generative AI

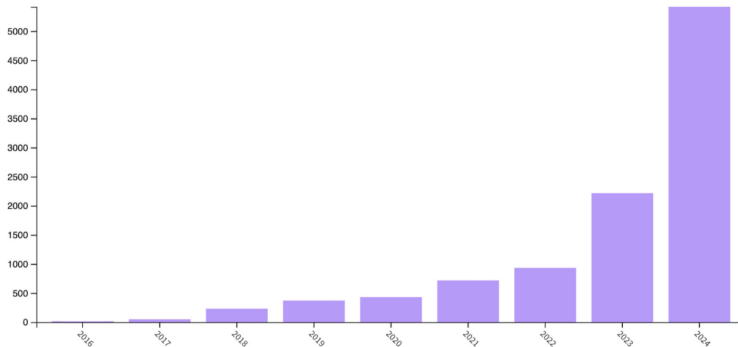


Figure: Web of Science search for “generative AI” (all fields)

Description of gen AI

- ▶ (Wiki) Generative artificial intelligence is a subset of artificial intelligence that uses generative models to produce text, images, videos, or other forms of data. These models **learn the underlying patterns and structures of their training data and use them to produce new data based on the input**, which often comes in the form of natural language prompts.
- ▶ (MIT news article) Generative AI can be thought of as a machine-learning model that is trained to create new data, rather than making a prediction about a specific dataset. **A generative AI system is one that learns to generate more objects that look like the data it was trained on.**

Basic idea of generation

- ▶ Gen AI needs **training data**.
- ▶ Gen AI produces **new data**.
- ▶ Gen AI produces data that **looks like training data**.

Our contribution

- ▶ We formalize generation as a **sequential two-player game** between an adversary and a generator. The formulation generalizes the notion of “language generation in the limit” from Kleinberg and Mullainathan [2024].
- ▶ We introduce two new paradigms of generation called **“uniform” and “non-uniform” generation** and provide their characterizations.
- ▶ We uncover fundamental **differences between generation and prediction**.

Generation model

- ▶ Formalize [Kleinberg and Mullainathan, 2024] into learning theory.
- ▶ $\mathcal{X}$ be an abstract countable instance space
- ▶ $\mathcal{H} \subset \{0, 1\}^{\mathcal{X}}$ be a collection of functions that maps instances to a binary label $\{0, 1\}$
- ▶ For every $h \in \mathcal{H}$, define $\text{supp}(h) = \{x \in \mathcal{X} : h(x) = 1\}$.

Generation model

- ▶ $\mathcal{X} = \mathbb{N}$.
- ▶ For each n , define h_n such that $\text{supp}(h_n) = \{n, 2n, 3n, \dots\}$
- ▶ $\mathcal{H} = \{h_n : n = 1, 2, \dots\}$

	1	2	3	4	5	6	7	8	...
h_1	1	2	3	4	5	6	7	8	...
h_2		2		4		6		8	...
h_3			3			6			
h_4				4				8	...

Figure: Example of a hypothesis Class

Generative model

- ▶ Instance space $\mathcal{X}$; Hypothesis class $\mathcal{H}$.
- ▶ Learner knows $\mathcal{H}$.

Game

Adversary chooses $h^* \in \mathcal{H}$ and an enumeration of $\text{supp}(h^*)$ (without learner's knowledge). At time $t = 1, 2, \dots$

- ▶ Adversary reveals $x_t \in \text{supp}(h^*)$
- ▶ Learner generates $\hat{x}_t \in \mathcal{X}$ such that $\hat{x}_t \notin \{x_1, \dots, x_t\}$

Goal

Learner “generates” successfully if there is a finite time t^* s.t. for all $t \geq t^*$, $\hat{x}_t \in \text{supp}(h^*)$ (assume $|\text{supp}(h^*)| = \infty$)

	1	2	3	4	5	6	7	8	...
h_1	1	2	3	4	5	6	7	8	...
h_2		2		4		6		8	...
h_3			3			6			
h_4				4				8	...

For example, if adversary picks up h_2 , and reveals 4, 8, 12, ... The learner can always generate a multiple of 4 to win.

Three variants of the model

- ▶ **Gen in the limit:** t^* can depend on h^* and the enumeration.
[Kleinberg and Mullainathan, 2024]
- ▶ **Non-uniform gen:** t^* can depend on h^* but not on the enumeration.
- ▶ **Uniform gen:** t^* depends only on $\mathcal{H}$ [Kleinberg and Mullainathan, 2024]

Uniform Gen $\Rightarrow$ Non-uniform Gen $\Rightarrow$ Gen in the limit

Key results

- ▶ Introduce Closure Dimension $C(\mathcal{H})$,
- ▶ Both uniform and non-uniform generation can be characterized by Closure Dimension,
- ▶ Closure Dimension is also related to gen in the limit.

Version space

- ▶ Suppose learner has observed $x_1, \dots, x_n$
- ▶ The “version space” is then

$$\mathcal{H}_{(x_{1:n})} := \{h \in \mathcal{H} : \{x_1, \dots, x_n\} \subseteq \text{supp}(h)\}$$

- ▶ If $x \in \text{supp}(h)$ for **every** h in the version space, it is safe to play it

Closure dimension

That is, it is safe to play from

$$\bigcap_{h \in \mathcal{H}_{(x_{1:n})}} \text{supp}(h)$$

Closure

$$\langle x_{1:n} \rangle_{\mathcal{H}} = \begin{cases} \perp & \text{if version space empty} \\ \bigcap_{h \in \mathcal{H}_{(x_{1:n})}} \text{supp}(h) & \text{otherwise} \end{cases}$$

Closure dimension

- $\mathcal{X} = \mathbb{N}$, $\text{supp}(h_n) = \{n, 2n, 3n, \dots\}$

Observed Instances	Version Space $\mathcal{H}_{(\cdot)}$	Closure $\langle \cdot \rangle_{\mathcal{H}}$
4	$\{h_1, h_2, h_4\}$	$\{4, 8, 12, \dots\}$
4, 6	$\{h_1, h_2\}$	$\{2, 4, 6, \dots\}$
4, 6, 5	$\{h_1\}$	$\mathbb{N}$

Closure dimension

- ▶ As $n \uparrow$, the version space shrinks and $\langle x_{1:n} \rangle_{\mathcal{H}}$ expands; once closure is infinite, learner can win
- ▶ Consider the smallest non-negative integer d such that, for **every** $x_1, \dots, x_{d+1}$, either the version space is empty or the closure is infinite
 - ▶ We call d the **closure dimension** of $\mathcal{H}$

Key results

- ▶ $\mathcal{H}$ is uniform gen $\Leftrightarrow C(\mathcal{H}) < \infty$
- ▶ Finite class has finite closure dimension.

Corollary

Any finite class is uniformly generatable [Kleinberg and Mullainathan, 2024].

Key results

- ▶ $\mathcal{H}$ is non-uniform gen $\Leftrightarrow$ There exists $\mathcal{H}_i, i = 1, 2, \dots$ with $C(\mathcal{H}_i) < \infty$ s.t. $\mathcal{H}_i \uparrow \mathcal{H}$

Corollary

Any countable class is non-uniformly generatable.

- ▶ Corollary strictly improves Kleinberg and Mullainathan [2024] first main result
- ▶ Independently proved by Charikar and Pabbaraju [2024]

Key results

- ▶ $\mathcal{H}$ is gen in the limit $\Leftrightarrow$ There exists $\mathcal{H}_i, i = 1, \dots, n$ with $C(\mathcal{H}_i) < \infty$ s.t. $\mathcal{H} = \bigcup_{i=1}^n \mathcal{H}_i$

Theorem

There exists a **countable sequence** $\mathcal{H}_1, \mathcal{H}_2, \dots$ such that $C(\mathcal{H}_i) = 0$ for all $i \in \mathbb{N}$ but $\bigcup \mathcal{H}_i$ is **not** generatable in the limit

Open Question

What characterizes generatability in the limit?

Generation vs Prediction

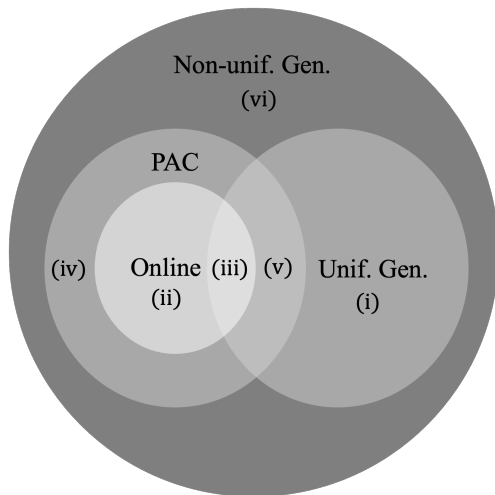


Figure: Prediction is incomparable with generation

Generation vs Prediction

- ▶ The best evidence of this difference is their behavior under unions.
- ▶ Prediction is well-behaved under unions: if $\mathcal{H}_1$ and $\mathcal{H}_2$ are PAC/online learnable then so is $\mathcal{H}_1 \cup \mathcal{H}_2$.
- ▶ This is not true for generation.

Lemma

There exists a class $\mathcal{H}$ and a hypothesis h such that $C(\mathcal{H}) = 0$ but $\mathcal{H} \cup \{h\}$ is not non-uniformly generatable.

More follow-up work in the K&M model

- ▶ We also study **prompted generation**.
- ▶ [Charikar and Pabbaraju, 2024] study **non-uniform generation, validity vs breadth** and **additional feedback**.
- ▶ [Kalavasis et al., 2024b] and [Kalavasis et al., 2024a] study **tradeoffs between validity and breadth**.
- ▶ [Raman and Raman, 2025] study generation with **noisy examples**.
- ▶ [Peale et al., 2025] require outputs are distribution that are **representative of input data**.

Future Directions

- ▶ **Stochastic Generatability:** what characterizes generatability when positive examples are drawn iid from a distribution?
- ▶ **Private Generatability:** how does generatability change if one requires generators to be differentially private?
- ▶ **Continuous Generatability:** What if the instance space is no longer a discrete space, (e.g., the Euclidean space), how could we define generatability?

Open problems

- ▶ What characterizes generatability in the limit?
- ▶ Is generatability in the limit closed under finite unions?
- ▶ Is the finite union of non-uniformly generatable classes generatable in the limit?
 - ▶ Solved by [Hanneke et al., 2025], the answer is no!

Theorem [Hanneke et al., 2025]

There exists classes $\mathcal{H}_1$ and $\mathcal{H}_2$ that are both non-uniformly generatable, but $\mathcal{H}_1 \cup \mathcal{H}_2$ is not generatable in the limit.

Thanks for listening!

References

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